XML Writing and Parsing

SOA – lab2



Agenda

- The Difference Between XML and HTML
- XML Usage
- Importance of XML
- XML main components
- How to write an XML file?
- XML parsing

The Difference Between XML and HTML

XML is not a replacement for HTML.

XML and HTML were designed with different goals:

XML was designed to describe data, with focus on what data is

HTML was designed to display data, with focus on how data looks

HTML is about displaying information, while XML is about carrying information.

XML is a software- and hardware-independent tool for carrying information.

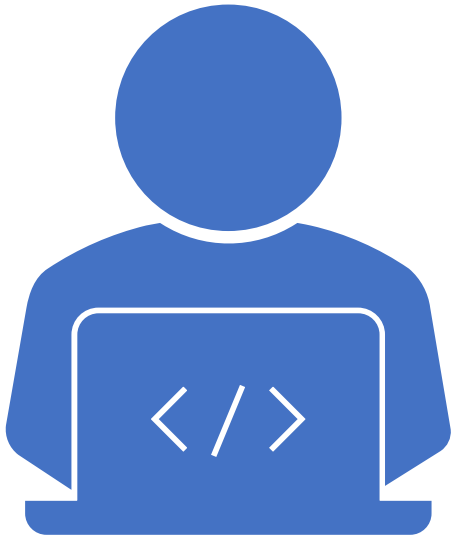


XML Usage

- XML can be used to exchange the information between organizations and systems.
- XML can be used for offloading and reloading of databases.
- XML can be used to store and arrange the data, which can customize your data handling needs.
- XML can easily be merged with style sheets to create almost any desired output.
- Virtually, any type of data can be expressed as an XML document.

Importance of XML

- **XML Separates Data from HTML**
- With XML, data can be stored in separate XML files. This way you can concentrate on using HTML/CSS for display and layout
- **XML Simplifies Data Sharing**
- In the real world, computer systems and databases contain data in incompatible formats.
- XML data is stored in plain text format. This provides a software- and hardware-independent way of storing data.
- This makes it much easier to create data that can be shared by different applications.
- **XML Simplifies Data Transport**
- One of the most time-consuming challenges for developers is to exchange data between incompatible systems over the Internet.
- Exchanging data as XML greatly reduces this complexity, since the data can be read by different incompatible applications.



Importance of XML Cont’d

- **XML Simplifies Platform Changes**
- Upgrading to new systems (hardware or software platforms), is always time consuming. Large amounts of data must be converted and incompatible data is often lost.
- XML data is stored in text format. This makes it easier to expand or upgrade to new operating systems, new applications, or new browsers, without losing data.
- **XML Makes Your Data More Available**
- Different applications can access your data, not only in HTML pages, but also from XML data sources.
- **Internet Languages Written in XML**
- Several Internet languages are written in XML.
- One of the beauties of XML, is that it can be extended without breaking applications

XML Main Components

- **XML Declaration :**
- The XML document can optionally have an XML declaration. It is written as follows:
- `<?xml version="1.0" encoding="UTF-8"?>`
- Where *version* is the XML version and *encoding* specifies the character encoding used in the document.
- **Tags and Elements**
- An XML file is structured by several XML-elements, also called XML-nodes or XML-tags. The names of XML-elements are enclosed in triangular brackets < > as shown below:
- `<element>`
- **Root Element:**

```
<root>
        <x>...</x>
        <y>...</y>
</root>
```

XML Main Components Cont’d

- **XML Attributes**
- An **attribute** specifies a single property for the element, using a name/value pair. An XML-element can have one or more attributes.
- For example:
- `<a href="http://www.tutorialspoint.com/">Tutorialspoint!</a>`
- Here ***href*** is the attribute name and ***http://www.tutorialspoint.com/*** is attribute value.

XML Attributes for Metadata

- Sometimes ID references are assigned to elements. These IDs can be used to identify XML elements in much the same way as the id attribute in HTML.
- <messages>

```
<note id="501">
    <to>Tove</to>
    <from>Jani</from>
    <heading>Reminder</heading>
    <body>Don't forget me this weekend!</body>
</note>
```

- The id attributes above are for identifying the different notes. It is not a part of the note itself.
- Metadata (data about data) should be stored as attributes, and the data itself should be stored as elements.

How to write an XML file?

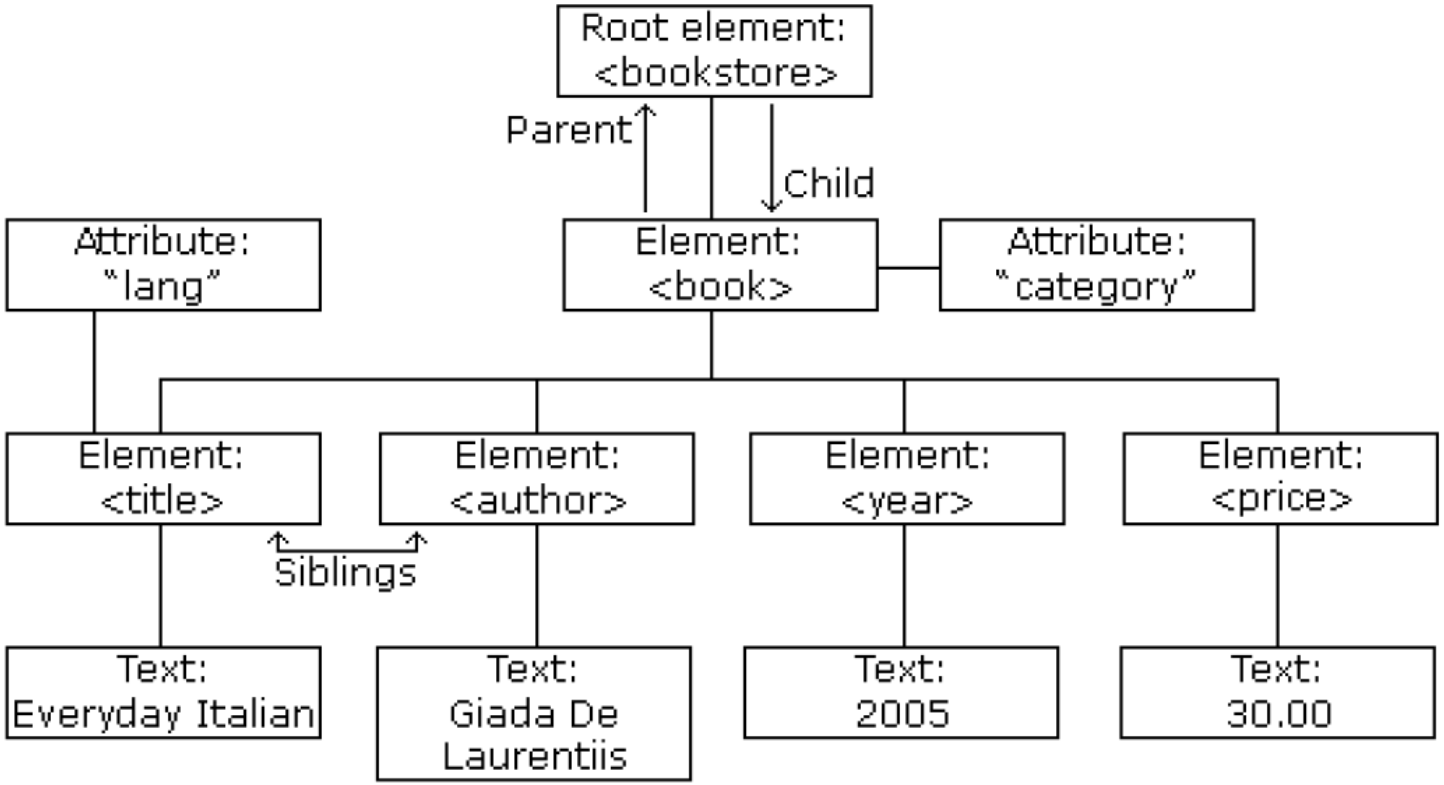
- The first line in an XML document is called the **prolog, it contains XML version, encoding**

```
<?xml version="1.0" encoding="UTF-8" ?>
```

- Define the **root** element of your file
- The **root** element is the first tag you will define in the XML file which will contain all other elements
- Define the **child** elements that you will have, the **content** of each, and **attributes** if any
- For Example:

```
<contact-info>
    <name>Tanmay Patil</name>
    <company>TutorialsPoint</company>
    <phone>(011) 123-4567</phone>
</contact-info>
```

XML Example


Root element:
<bookstore>
Parent
Child
Attribute:
"lang"
Element:
<book>
Attribute:
"category"
Element:
<title>
Element:
<author>
Element:
<year>
Element:
<price>
Siblings
Text:
Everyday Italian
Text:
Giada De
Laurentiis
Text:
2005
Text:
30.00

XML Example Cont’d

```
<bookstore>
        <book category="COOKING">
                <title lang="en">Everyday Italian</title>
                <author>Giada De Laurentiis</author>
                <year>2005</year>
                <price>30.00</price>
        </book>
        <!-- This is a comment -->
        <book category="CHILDREN">
                 <title lang="en">Harry Potter</title>
                <author>J K. Rowling</author>
                <year>2005</year>
                <price>29.99</price>
        </book>
</bookstore>
```

Parse XML in Java

- Use **DOM (Document Object Model) API** provides the classes to read and write an XML file.
- DOM parser parses the entire XML file and creates a **DOM** object in the memory.
- It models an XML file in a **tree structure** for easy traversal and manipulation.
- In DOM everything in an XML file is a **node**.

Parse XML in Java Cont’d

- **Instantiate XML file:** DOM parser loads the XML file into memory and consider every tag as an element.
- **Get root node:** Document class provides the **getDocumentElement()** method to get the root node and the element of the XML file.
- **Get all nodes:** The **getElementByTagName()** method retrieves all the specific tag name from the XML file.
- Where **ELEMENT_NODE** type refers to a non-text node which has sub-elements.
- If we need to access all nodes from the starting, including the root node, we can recursively call the getChildElement() method.

Parse XML in Java Cont’d

- **Get Node by text value:** We can use **getElementByTextValue()** method in order to search for a node by its value.
- **Get Node by attribute value:** If we want to search a node by the value of a specific attribute, we can use the getElementByTagName() method along with getAttribute() method.

References

- https://developer.mozilla.org/en-US/docs/Web/XML/XML_introduction
- https://www.javatpoint.com/how-to-read-xml-file-in-java
- https://www.journaldev.com/848/java-file-path-absolute-canonical
- https://www.tutorialspoint.com/java_xml/java_dom_parse_document.htm